

Overhead Crane Safety Standard

CLEVELAND-CLIFFS STEEL
Overhead Crane Safety Standard

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1 Purpose/Scope

- 1.1 This policy establishes minimum standards for activities that support the inspection, operation, maintenance, boarding/de-boarding, and other safe work practices related to crane operations.
- 1.2 This policy applies to employees of Cleveland-Cliffs Steel business units, contractors, subcontractors and hired labor employees performing work for or on behalf of Cleveland-Cliffs Steel on premises owned, rented, or otherwise occupied by Cleveland-Cliffs Steel.
- 1.3 The policy applies to Electric Overhead Traveling (EOT) gantry, and semi-gantry cranes including cab, pendent and radio control operated cranes.

2 Responsibilities and Authorities

- 2.1 This standard was jointly developed with participation from the represented workforce and management.



2.2 Qualified Crane Operators Are Responsible For:

- Attending crane operator training and following the requirements specified in this standard.
- Performing pre-use inspections to ensure cranes are operating properly and to verify that cranes can be operated safely.
- Immediately reporting any critical defects or other operational conditions that may affect the safe operation of the crane.
- Obeying verbal instructions or hand signals provided by designated shop floor personnel to ensure lifts and crane movements are conducted in a safe manner that avoids potential impact to equipment, personnel, or other objects.
- Safely operating cranes and avoiding movements or unsafe practices that can jeopardize the safety of personnel or risk equipment or property damage.
- Keeping crane cabs free of clutter and performing necessary housekeeping.
- Avoiding distractions or performing other tasks that may take the operators attention away from critical actions required to safely operate the crane.
- Using horns or other audible devices to alert ground personnel of the movement of the crane.
- Immediately reporting any crane related incidents including personal injuries, near misses or equipment damage events.

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2.3 Qualified Ground Personnel Are Responsible For:

- Maintaining awareness of crane operations in their work area and verify safe approach when entering buildings, crossing medians, exiting pulpits or working around crane lifts or other areas where crane operations are conducted.
- Receiving approval from the Qualified Crane Operator and verifying conditions to ensure the safe boarding of an occupied crane.
- Ensuring appropriate and recognized hand signals or approved radio communications are used to ensure coordination with Qualified Crane Operators.
- Suspending crane operations when an unsafe or at-risk condition is observed.
- Using sound rigging practices and when necessary, performing low/limited lifts to ensure load stability or to verify the center of gravity of a load.
- Using no-touch tools when engaging/disengaging safety latches or when manipulated rigging hardware or lifting cables that are under tension.
- Using tag lines or other approved no-touch tools to provide a safe distance when stabilizing, moving, or adjusting suspended loads.
- Clearing non-essential personnel from an area where a lift is to be performed and keeping hands and other body parts away from potential pinch points or other hazards associated with crane movements and lifts.

3 References and Definitions

3.1 References

- MIOSHA General Industry R 408.12004
- OSHA 29 CFR 1910.179, Overhead and Gantry Cranes
- 29 CFR 1910.181, Derricks
- American Society of Mechanical Engineers. (2005). ASME B30.10: Hooks
- American Society of Mechanical Engineers. (2006). ASME B30.20: Below-the-Hook Lifting Devices
- American Society of Mechanical Engineers. (2016). ASME B30.2: Overhead and Gantry Cranes
- MIOSHA-STD-1116 Part 18: Overhead and Gantry Cranes.

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3.2 Definitions

Amber flashing lights: Rotating or blinking warning lights visible to the crane operators shall be used to draw attention to temporary work platforms, ladders or mobile equipment erected or staged that do not impinge on the crane envelope. This does not apply to normal storage of ladders in the upright position or wooden access platforms that provide access to equipment such as stairs or decking. A crane shall proceed with caution past a flashing amber light maintaining maximum distance as possible with lifting devices and suspended loads.

Authorized employee: An employee that has been trained on the applicable procedure and has been classified as "authorized" by management.

Blue flashing lights: Rotating or blinking warning lights visible to the crane operators shall be used to draw attention to a planned work zone. A crane shall never go past a flashing blue light.

Bridge: The part of a crane consisting of girders, trucks, end ties, foot walks and the driving mechanism which carries the trolley or trolleys.

Competent Person, Trained and Competent. (Crane Interference Form trained and Cleveland-Cliffs steel management or designees): A person who has the necessary experience of the crane and equipment used in the lifting operation to carry out the function satisfactorily, who is capable of identifying an existing or potential hazard in surroundings, or under working conditions, that are hazardous or dangerous to an employee, and who has the authority and knowledge to initiate prompt corrective measures to eliminate the hazards and implement the Crane Interference Form (CCS_HHS_FRM_14.1).

Crane – Cab-Operated Crane: A crane controlled by an operator in a cab located on the bridge or trolley.

Crane – EOT Cranes (Electric Overhead Traveling): A crane with a movable bridge carrying a movable or fixed hoisting mechanism and traveling on an overhead fixed runway structure.

Crane – Gantry Crane: A crane like an overhead crane except that the bridge for carrying the trolley or trolleys is rigidly supported on two or more legs running on fixed rails or another runway.

Crane – Jib Cranes (Wall Crane): A crane having a jib with or without trolley and supported from a side wall or line of columns of a building. It is a traveling type and operates on a runway attached to the side wall or columns.

Crane (cab operated, remote operated, pendant operated and automatic cranes, EOT Crane): Includes bridge, trolley, end trucks, cab and if applicable, permanently attached (wire rope reeved) lifting devices (block, Heppenstall, j-hooks, etc.).

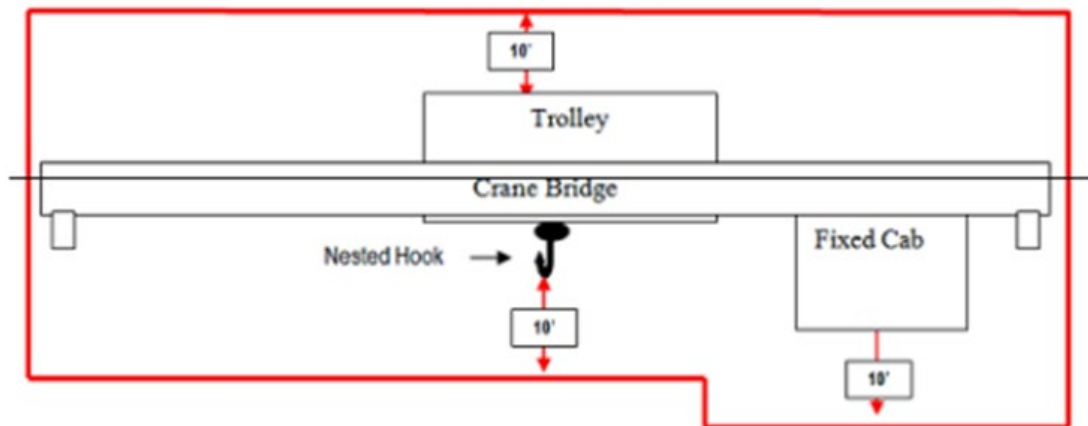
Crane – Automatic Crane: A crane which when activated operates through a preset cycle or cycles.

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Crane Envelope: The following are the conditions defining the crane envelope:

- Ten (10) feet above the trolley.
- Ten (10) feet directly below and adjacent to a fixed cab.
- Ten (10) feet below the bridge, unless one of the following conditions exists:
 - a. Crane cab is mounted on the trolley, then it is ten (10) feet below the cab and across the length of the bridge.
 - b. Ten (10) feet below the lowest part of the nested lifting device and across the length of the bridge. For example: if a set of j-hooks when nested extend 15 feet below the bridge, the envelope is 25 feet below the entire length of the bridge.

Crane Envelope for a Fixed Cab Overhead Crane with a Nested Hook that Extends Below the Crane Bridge



Crane Lift Plan or Lifting Plan: Provides a systematic approach towards planning a lift and ensures that the lifting operation is properly managed. For example, a lifting plan would ensure that the crane is safely deployed and operated, and the lifting crew are clear about their roles and responsibilities.



Critical Lift: A lift that (1) exceeds 75 percent of the rated capacity of the crane or derrick, or (2) requires the use of more than one crane or derrick. *Critical Lifts, Abnormal Lifts, and High-Risk Lifts need a Crane Lift Plan. Refer to section 5.19.*

Designated Safe Location (applies only to planned work zones): Workers must be protected from being struck by other cranes when permission is given by safety lookouts to move through a planned work zone. The group must be positioned so that they may not be injured by movement of other cranes into or through the planned work zone when permission is given for such movement. The designated safe location may be within the crane envelope, but the crane, crane cab or the permanently attached lifting device in the nested position shall not be able strike the workers.

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Extended Point: For the extent of this document, it would be the reaching point of a fully extended boom of a mobile crane or the fully extended reach of a man lift.

Hot Rail: Electrical conductors (runway conductors) located along a crane runway to provide power to the crane.

Impinge: To encroach, infringe on the crane envelope, to influence the crane.

Impinging – Capable of Impinging: To be capable of entering into the crane envelope. Example: a 90-foot man-lift working in a building that is 100 ft. tall is capable of impinging into the crane envelope, even if it is only being used at a height of 40 feet.

Limit Switch: A switch which is operated by some part or motion of a power-driven machine or equipment to alter the electric circuit associated with the machine or equipment.

Load: Refers to the object(s) being hoisted and/or the weight of the object(s); both uses refer to the object(s) and the load-attaching equipment, such as, the load block, ropes, slings, shackles, and any other ancillary attachment.

Load Block (Crane Block): The load block is the assembly of hook or shackle, swivel, bearing, sheaves, pins, and frame suspended by the hoisting rope.

Monorail System (hoist, crane): An overhead track from which hoisting equipment travels to transport loads and includes curves, switches, transfer devices, drop sections, hangers, and related equipment.

Multiple Crane Lifts: Those which involve two or more cranes to support a load either with hooks attached directly to the load by slings, through shackles attached to pad eyes or directly to an equalizer beam.

Programmed Work Zone: Programmed work zone - computer program restrictions by the department and/or automation engineers that sets crane(s) operational boundaries for automatic cranes.

Qualified Crane Operator: A person who has been trained in the operation, function and testing of the crane that has successfully demonstrated the ability to operate the crane and has been deemed qualified and assigned by management as the principal operator of the crane.

Qualified Floor Person: A person who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge or training and experience, successfully demonstrated the ability to solve/resolve problems related to the subject matter, the work, or the project.

Rail Stop (rail dog): A device attached to a crane bridge rail or fixed structure to limit the travel of the crane bridge.

Safety Latch: A hook with a latch to prevent slings or load from accidentally slipping off the hook.

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Safety Lookout: A person whose sole responsibility is to look out for the crew or crews being protected and to prohibit the crane operator (one Lookout per crane) from operating the crane too close to the personnel being protected. The safety lookout is assigned to the group doing the repair or maintenance activity. This person reports to and is relieved with the work group. The safety lookout may be positioned in the crane cab or on the ground with the work crew based on the circumstances of the work to be done. One safety look out will be assigned to each crane on either side of the work area.

Side Pull, Side Swing: The portion of the hoist pull acting horizontally when the hoist lines are not operated vertically.

Suspended Load: Anything that is lifted above ground level.

Trolley: The unit which travels on the bridge rails and carries the hoisting mechanism.

Work Group Supervisor or Leader: The individual in charge of the work that is to be performed. In most cases this is not the operations supervisor but the repair group supervisor doing the work activity.

4 Resources

4.1 None

5 Requirements

5.1 Medical Qualifications

- 5.1.1 Only trained and authorized personnel are allowed to operate a crane. Qualification as a crane operator shall be based on medical certification, training and demonstrated ability to operate any crane to which the employee may be assigned. Cranes shall be operated in a competent manner in strict compliance with government regulations and all applicable crane safety rules.
- 5.1.2 All crane operators must be deemed medically qualified to operate any crane.

5.2 General Crane Awareness Training

- 5.2.1 All employees shall be trained at least to an awareness level on crane safety and the requirements within this standard.
- 5.2.2 Cleveland-Cliffs Steel managers and designees, affected employees shall be trained initially prior to assignment and whenever there is a significant change in this standard and shall be trained in the following:

5.2.2.1 Crane Interference Form requirements:

- How to properly complete and implement Crane Interference Form (CCS_HHS_FRM_14.1).

- 5.2.3 Safety lookouts shall be trained in the following:

- The location and boundaries of the work area.
- Specific responsibilities, duties, and obligations to the work crew.
- The signs and signals of the job.
- Responsibility of not leaving the job until relieved by the work crew supervisor or leader.

5.3 Initial Training

- 5.3.1 All newly assigned crane operators must attend formal basic skills training including classroom and hands-on training. Each crane operator must successfully complete a defined period of on-the-job training and a performance evaluation under the direction of designated approved trainers prior to being qualified as a crane operator.



- 5.3.2 *The successful completion of training shall include documentation (e.g., quiz or post-training test) that the crane operator has acquired the understanding, knowledge, and skills necessary to safely perform their duties.*

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5.3.3 Training shall include:

- The operator's responsibility is to perform and document a pre-use inspection of the crane.
- Basic design and operating functions of cranes.
- Limit switch testing.
- Proper execution of each specific lifting procedure, including:
 - a. Environmental hazards – weather
 - b. Pre-use inspections/tests
 - c. Load weight estimation
 - d. Emergency procedures
 - e. Traveling with load control
 - f. Rigging
 - g. Hand signals
 - h. Load dynamics
 - i. Limit switches
 - j. Warning signals
 - k. *How to properly complete and implement the Crane Interference Form (CCS_HHS_FRM_14.1).*
 - l. Responsibilities of the safety lookout

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5.3.4 Refresher training must be provided under the following circumstances:

- Observation of unsafe operation of the crane.
- Operator was involved in an accident or a near-miss incident.
- Conditions in the workplace or equipment change in a manner that could affect the safe operation of the crane.
- An extended period in which the operator has not operated a crane and feels they need refresher training.

Note: the exact content and duration of refresher training will be specific to and consistent with the circumstances above. What is necessary to close any training and operating skills gaps is the question that must be satisfied.

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- 5.3.5 The successful completion of training shall include documentation (e.g., quiz or post-training test) that the crane operator has acquired the understanding, knowledge, and skills necessary to safely perform their duties.
- 5.3.6 A documented crane evaluation form shall also be used to document completion of a crane operator's on-the-job training and their ability to safely perform the required crane operations associated with their crane assignment.
- 5.3.7 Rigging training must include the following:
 - a. Select tackle and hardware suitable to the load.
 - b. Environmental hazards – weather.
 - c. Inspection/tests.
 - d. Load weight estimation.
 - e. Rigging equipment selection.
 - f. Hand signals.
 - g. Maintenance/storage of slings and rigging components.
 - h. Rigging/operating safe practices.
 - i. Load-indicating devices.
 - j. Personal protective equipment.

5.4 Training Documentation

- 5.4.1 All training must be documented as either initial or refresher and maintained for the period that all other training records are maintained (both immediately accessible or via archives) per each business unit's practice.

5.5 Crane Boarding/De-Boarding

- 5.5.1 Crane boarding that involves a host department and a service department employee or crew must be coordinated so that the host department is aware of the activity and can provide further guidance on any issues/scheduling of the crane in question. Full and on-going communications is required.
- 5.5.2 Fall protection requirements both fixed and personal must be discussed and implemented.
- 5.5.3 No one shall ever board or de-board a moving crane.
- 5.5.4 Personnel, other than qualified crane operators, shall not be permitted on cranes, crane runways or in crane cabs without the authorization of management.

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5.5.5 Verify status lights before boarding any crane. Status lights shall be visible from the boarding platform.

 5.5.6 *Sites can establish and train personnel on the site's standard for crane status light colors.*

 5.5.7 *The site's standard configuration for crane power and magnet power is arranged to include a fail-safe so that the status can be determined if a light is not functioning. (e.g., magnet indicator light is off to indicate power is on.)*

5.5.8 A documented procedure for boarding and de-boarding occupied and unoccupied overhead cranes shall be developed and implemented. The procedure shall have, at minimum, the following instruction:

5.6 Boarding a Crane That is Occupied

5.6.1 The authorized employee shall notify the crane operator of their intention to board the crane and shall receive acknowledgement from the crane operator.

5.6.2 The crane operator shall de-energize the crane.

5.6.3 The crane operator shall assure the crane power OFF indicator status light is lit and then place the OK to board indicator light to ON if the crane is so equipped.

5.6.4 The crane operator shall then exit the cab and go to a position where they can be seen by the employee(s) wishing to board the crane.

5.6.5 The crane operator may put the crane into operation only after ensuring all employees are clear, the Crane Power ON light is turned on and sounds the crane warning for at least five seconds.

5.7 Boarding or De-Boarding a Crane During Planned Maintenance Downturns and a Crane Operator is Not Scheduled

5.7.1 If the hot rails are locked out and/or a lockout box established, the employee can enter the crane after putting their personal lock on the hot rail source or lockbox for the hot rails and signing the lockout control sheet.

5.7.2 If the hot rails are not locked out, the employee shall gain permission by the supervisor in charge and/or the maintenance crew leader prior to boarding the crane.

5.7.3 The crane must be de-energized prior to boarding the crane.

5.7.4 The manual magnetic switch must be in the off or open position and the indicator light on before the crane can be boarded or exited.

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5.7.5 Signs shall be posted at the bottom of the crane access stairs or ladders and at each boarding location stating, at a minimum, the following wording:

- Warning – overhead crane boarding location. Authorized employees only. Do not proceed unless trained and follow the requirements of the procedure.

5.8 Pendant Controlled Crane – Boarding

5.8.1 Boarding must not occur until the operator and/or supervision has been notified.

5.8.2 The operator or a designated person must be stationed at the controls while boarding or de-boarding occurs.

5.9 Radio Controlled Crane – Boarding

5.9.1 The boarding of radio-controlled cranes must include the following minimum requirements:

- a. No one is to board radio-controlled equipment until the operator is notified of the intent to board. In the event there is no operator, contact the area supervisor.
- b. Secure both the radio control box (transmitter) and key making sure all are in the “off” position and key is removed from transmitter and take them with you to the crane.
- c. Turn off (open) the boarding switch, if applicable. Even if a spare r/c transmitter is put into service, the crane will not operate from the radio control mode.
- d. Look inside the crane cab to make sure that an operator is not present. The crane can be operated manually (from the cab) even if the board switch is off. The boarding switch only effects radio control power.
- e. Board the crane. If no one is in the crane cab, it is safe to board. If someone is in the crane cab, they must turn off the cab power switch and leave the cab before it is safe to board.
- f. Use both hands when climbing the boarding ladder.
- g. Put the R/C transmitter in a place where it will not fall from the crane or be in the way.
- h. Turn off (open) the main power switch.
- i. Stand to the right of the main power switch, turn your face away and turn off the switch with your left hand.
- j. After de-boarding the equipment, those contacted prior to boarding (operator/supervision) must again be notified.

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5.10 Inspections

5.10.1 Pre-use Inspection

5.10.1.1 Each crane operator shall complete the Pre-use Inspection prior to initiating the first lift of their shift.

- Exception: if a lift is required immediately at the start of the shift, the upper control limit (s) must be checked prior to initiating the lift. As soon as that lift is completed the remainder of the Pre-use Inspection form shall be completed.

△ 5.10.1.2 OSHA: 29 CFR 1910.179 (n) (4) (i):
*"At the beginning of each operator's shift, **the upper limit switch of each hoist shall be tried out under no load. Extreme care shall be exercised; the block shall be 'inched' into the limit or run in at slow speed.**"*

△ 5.10.1.3 ASME B30.2-2011-2.1.3 :
*"The primary upper limit device **under a no-load condition** with the block either running at slow speed or inched to the limit."*

△ 5.10.1.4 *Lifting devices that are not reeved onto the hoist ropes, such as hook-on buckets, magnets, grabs, and other supplemental devices used for ease of handling certain types of loads. The weight of these devices is to be considered part of the rated load.*

5.10.2 During the Pre-use Inspection, if any deficiencies are noted that affect the safe operation of the crane:

- a. Notify the operation Shift Manager/Crew Lead immediately.
- b. The crane shall be parked at nearest boarding location and the crane operator shall de-energized the crane.
- c. The operation Shift Manager/Crew Lead shall notify maintenance immediately.
- d. Turn the signed Pre-use Inspection form into the operation Shift Manager/Crew Lead by the end of the shift.
- e. The operation Shift Manager/Crew Lead shall review, sign the Pre-use Inspection, and forward it to the section manager by the end of the turn. If a non-safety related item is noted on the Pre-use Inspection form, the section manager or designee shall issue a work order to correct the issue.

5.10.3 During the Pre-use Inspection if no deficiencies are noted:

- a. Complete and sign the Pre-use Inspection form as soon as the inspection is complete.

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- b. Turn the signed Pre-use Inspection form into the operation Shift Manager/Crew Lead at end of the shift.
- c. The operation Shift Manager/Crew Lead shall review, sign the Pre-use Inspection form, and forward it to the Section Manager prior to the end of the shift.
- d. The Section Manager (or equivalent) must review and sign each completed crane pre-use inspection sheet by the first workday after the date of inspection. If a non-safety related item is noted on the Pre-use Inspection form, the Section Manager or designee shall issue a work order to correct the issue.

5.10.4 Pre-use Inspection forms shall be filed and kept for at least six (6) months.

Δ 5.11 Frequent, Periodic, Comprehensive and Other Inspections

5.11.1 *Frequent and periodic inspections shall be performed in accordance with OSHA (Occupational Safety & Health Administration) and manufacturer requirements.*

5.11.2 *Monthly Inspection*

5.11.3 *A standard checklist must be used to document the inspection and track correction of deficiencies as outlined under Crane Operator Pre-use Inspection requirements above.*

5.11.4 *All Components listed under Crane Operator Pre-use Inspection requirements must be inspected.*

5.11.5 *Additionally, inspect for:*

- a. *Deformed, cracked, or corroded structural members.*
- b. *Loose bolts or rivets.*
- c. *Cracked or worn sheaves and drums.*
- d. *Hoist Cables.*
- e. *Worn, cracked, or distorted parts, such as pins, bearings, shafts, gears, rollers, locking or clamping devices.*
- f. *Excessive wear to brake system parts, linings, pawls, and ratchets.*
- g. *Load wind and other indicating devices.*
- h. *Magnet attachment connections if applicable.*

5.11.6 *Corrective action of any noted deficiencies must be tracked and audited for timely correction.*

5.11.7 *Monthly Hooks, Lifting Devices and Hoist Cables Inspection.*

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- 5.11.8 *A visual inspection of the hooks, lifting devices and hoist cables must be conducted at least monthly.*
- 5.11.9 *The inspection must be documented, including the inspection date, signature of the person conducting the inspection and serial number or identifier of the item(s) inspected.*
- 5.11.10 *The inspection reports must be retained for a period of 6 months or until the deficiencies are corrected.*
- 5.11.11 *The monthly hook, lifting device and hoist cable inspection reports must be retained for a period of 6 months.*
- 5.11.12 *The monthly inspection reports must be retained for a period of 6 months.*

Note: The Monthly Inspection can be incorporated into the Annual inspection.

Δ 5.12 Annual Inspection

- 5.12.1 *A detailed inspection of each crane in regular service must be conducted at least annually. Refer to CCI Corporate Infrastructure Procedure # 2.10, Cranes – Structural Inspection Procedure.*
- 5.12.2 *These inspections must be documented.*
- 5.12.3 *Items and conditions to inspect are:*
 - a. *All requirements of Frequent Inspection.*
 - b. *Deformed, cracked, or corroded members.*
 - c. *Loose bolts or rivets.*
 - d. *Cracked or worn sheaves and drums.*
 - e. *Worn, cracked or distorted parts such as pins, bearings, shafts, gears rollers, locking and clamping devices.*
 - f. *Excessive wear on brake system parts, linings, pawls and ratchets.*
 - g. *Load, wind, and other indicators over their full range for any significant inaccuracies.*
 - h. *Excessive wear of chain drive sprockets and excessive chain stretch.*
 - i. *Electrical apparatus, for signs of pitting or any deterioration of controller contactors, limit switches and pushbutton stations.*
- 5.12.4 *The annual structural inspection reports must be retained for a period of 2 years.*

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△ 5.13 Cranes Not in Regular Use

- 5.13.1 *A crane which has been idle for a period of one month or more, but less than six months, an inspection must be conducted as described in 2.2 and 2.4 before placing into service.*
- 5.13.2 *A crane which has been idle for a period of over six months, an inspection must be conducted, as described in 2.2, 2.3 and 2.4 before placing into service.*
- 5.13.3 *Standby cranes must be inspected at least semi-annually, as described in 2.2 and 2.4, before placing into service.*

5.14 General Crane Rules

- 5.14.1 Crane operators shall not make or complete a lift where:
 - a. Line of sight to the load or lifting device can-not be maintained, unless directed by authorized personnel.
 - b. Signals shall be taken only from those authorized to give them.
 - c. Only one person shall be authorized to give signals for a given lift.
 - d. Crane operators must not make a lift, move, or place their load in any way, regardless of signals, if their judgment indicates someone is in a hazardous position or it is an unsafe lift.
- 5.14.2 Crane operators shall not lower or set a lift if floor personnel are able to be struck or caught by the lifting or lift device.
- 5.14.3 Crane operators shall not relax tension on a lift upon completing a lift if in their judgement, the lift is unstable or is in some other way creating a hazard.
- 5.14.4 Except for designated crane operations which are handled without a floor person or signal person in accordance with department procedures, a crane operator must not move a lift without standard hand signals or approved radio communication from a designated person (see appendix 2 for standard crane hand signals).
- 5.14.5 Do not lean lifts or lifting devices against surfaces or other objects without first providing some means to secure them from movement.
- 5.14.6 Unattended overhead cranes must be parked in a manner that all block(s) are parked no more than 2 feet above the floor or in a manner that restricts employees from being able to access the hazardous area(s) below all block(s).
- 5.14.7 Never carry any type of load over personnel.
- 5.14.8 A crane operator must always verify the positive engagement of crane j hooks or bales on ladle trunnions for all ladle lifts.

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- 5.14.9 The crane operator shall not hoist a hot metal lift until confirmation of positive hook up is done. This can be achieved through visual line of sight; camera systems or 3-part communication from a spotter.
- 5.14.10 Workers are not permitted to be inside a railcar while being loaded or unloaded.
- 5.14.11 Workers shall not be permitted on the bed of trucks being loaded or unloaded by a crane except for hooking and unhooking a lift. However, when loading, once the lift is stationary and no more than 6 inches above the blocking, the worker is permitted on the truck bed if necessary for final placement of the lift.
- 5.14.12 When loading and unloading a truck by crane, ensure the driver remains out of the cab and in the clear of the truck.
- 5.14.13 Maintain distance and control movement to avoid damage to railroad cars and trucks. Pay particular attention to safety grab irons, steps, and brake wheels. Report any damage to your supervisor.
- 5.14.14 Overhead cranes shall not be used for side pulls except when specifically authorized by Cleveland-Cliffs Steel Management and then only after all alternatives have been investigated.
- 5.14.15 A procedure shall be written for each side pull situation and will include: securing the load and hoist, evacuation of the area, conditions to stop the side pull and an inspection of the crane and cables immediately upon completion of the side pull.
- 5.14.16 The side pull will not be made if the responsible supervisor determines that the crane and/or the lifting mechanism will be over stressed.
- 5.14.17 Overhead cranes shall not be used to move railroad cars.
- 5.14.18 Crane operators must not hoist loads exceeding the safe load limit of their cranes. All EOT cranes, monorail hoists, jib cranes, crane blocks and lifting devices must have their lifting capacities clearly marked in a location visible to the operator and floor personnel.
- 5.14.19 The weight of the load plus the weight of the lifting device must be combined and be less than the capacity of the crane.
- 5.14.20 Crane operators shall sound the crane signal at the initial movements of the trolley, bridge and hoist when lowering.
- 5.14.21 Crane operators shall sound the crane signal when approaching any personnel in the area of the crane's travel.
- 5.14.22 Never carry any type of load over personnel. Avoid carrying loads over operating equipment unless when necessary to service the equipment or when there is no acceptable alternate route for the lift.
- 5.14.23 If manually activating the siren, activate it for at least 3 seconds and no more than 10 seconds for initial movement of the trolley, bridge and hoist when lowering.

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- 5.14.24 Suspended loads including any below-the-hook lifting devices (Heppenstalls, tongs, magnets, chains, etc.) shall not be left unattended. Crane operators shall not leave their position at the crane controls while the load is suspended.
- 5.14.25 Crane operators shall not engage in any activity that might divert their attention from a suspended load.

5.15 Pushing a Crane

- 5.15.1 Must be authorized by supervisor or designee.
- 5.15.2 Operators of adjacent cranes must be notified.
- 5.15.3 Cannot be pushed until it has been determined that no one is on the crane (other than in cab) or is attempting to board it.
- 5.15.4 A ground safety person is assigned to alert personnel to the activity above.
- 5.15.5 Pushing procedures must be developed and written. The following are example steps.
 - a. Approach but do not touch the crane.
 - b. Sound horn or other alarm device.
 - c. Touch and alarm at the same time as contact is made.
 - d. Push slowly to crane boarding location if operations permit.

5.16 Safe Operation with Lifting Magnets

- 5.16.1 Departmental safety rules shall identify specifically the type of lifts that may not be handled with electromagnets.
- 5.16.2 Do not carry magnet loads over machinery unless such movement has been approved by supervision.
- 5.16.3 Never carry lifts held by magnets near ground personnel or equipment.
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 5.16.4 *People with pacemakers are not to be working in close proximity to strong magnetic fields. Distances and precautions are to be determined by the Medical Department.*
- 5.16.5 Magnet lights indicate the status of the magnet power switch and must be maintained.

5.17 Connecting Magnets

- 5.17.1 Crane operator must lower and stop block/hook close to the ground.

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- 5.17.2 Crane operator is to place the magnet controller in the drop (off) position.
- 5.17.3 To ensure that the crane will not move, turn off the crane power.
- 5.17.4 Additionally: pull (open) the crane main mag power switch.
- 5.17.5 Crane operator is to notify ground person that power is off. (Ground person is to make certain the magnet signal light indicates power is off).
- 5.17.6 After verification and with the magnet light indicating power is off, the ground person is to place magnet chain on hook and plug in the connectors.
- 5.17.7 Crane operator hoists magnet by ground person's hand signal after all personnel are in the clear.

5.18 Disconnecting Magnets

- 5.18.1 Prior to disconnecting power to magnet, ensure that no load is on the magnet.
- 5.18.2 Crane operator must lower magnet to the ground.
- 5.18.3 Crane operator is to place the magnet controller in the drop (off) position.
- 5.18.4 With the controller in the drop (off) position and wait 10 seconds for power to drain.
- 5.18.5 Crane operator is to notify ground person that power is off. (Ground person is to make certain the magnet signal light indicates power is off.)
- 5.18.6 With ground person in the clear, crane operator is to lower magnet by hand signal from ground person.
- 5.18.7 To ensure that the crane will not move, turn off the crane power.
 - Additionally: pull (open) the crane main mag power switch.
- 5.18.8 After verification and with the magnet light indicates power is off, the ground person can now disconnect the chain, unplug the connector, and remove the magnet lifting ring off the hook.
- 5.18.9 With all personnel in the clear, crane operator can raise hoist on hand signal from ground person.

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5.19 Overhead Crane Lift Planning

5.19.1 The level of planning required for lifting is determined by the category of the lift.

Category of Lift	Examples (but not limited to)	Planning Required
High Risk / Abnormal Lift	• All multiple crane lifts. • Lifts over operating areas that may endanger personnel. • Lifts involving personal platforms (man baskets). • Lifts over or exceeding maximum rated loads.	• Risk Assessment • Develop a lift plan that addresses associated hazards. • Include crane operators and rigging crews in the lift plan development. A record of this involvement must be kept. • See below for details on what must be included in the lift plan
Standard Lifts with Standard Operating Procedure	• Standard required lifts throughout the plant.	• Risk assessment: Develop a standard operating procedure

5.19.2 The lift plan must include:

- a. Lift data: equipment weight, rigging weight, total weight, height of lift, radius of lift and equipment surface area, center of gravity.
- b. Equipment data: manufacturer, model, size, length block, material size.
- c. Rigging data: sling diameter, length, sling configuration, capacity, hook type, shackle size and capacity.
- d. Proximity to power lines, pipe racks and process areas.
- e. Local hazards and their controls and agreed communication method.

5.20 Lifting Devices

5.20.1 Qualified floor personnel must visually inspect below the hook lifting equipment such as chains, cables, slings, tongs, forks, etc., before use to ensure they are in good condition. Never use a defective lifting device or overload a lifting device.

5.20.2 Lifting devices and slings with defective or missing safety latches shall not be used.

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- 5.20.3 Lifting devices shall have permanently affixed and legible identification markings as prescribed by the manufacturer that indicate the recommended safe working load.
- 5.20.4 Lifting devices need to be certified (semi-annually) and identified with legible markings. Defective or out of date devices shall be removed from service, tagged, and reported to supervision immediately. Cable and sling eyes shall be cut and the device scrapped when found to be defective.
- 5.20.5 Inspect semi-annually
 - Non-Destructive Testing shall consist of "Magnetic Particle" and "Ultrasonic Testing" as the preferred methods. The "Die- Penetrant" and "Fluorescent Penetrant" methods may be used on non-magnetic material or material in which an induced magnetic field could cause damage, such as electric motors and armatures.
- 5.20.6 Two or more slings (wire, chain, rope or synthetic) shall not be connected to each other by means of choking, knotting, with shackles or other unapproved mechanical means. The proper length slings shall be used.
- 5.20.7 No lift shall be made with chains, cables, or slings while a magnet or other auxiliary lifting device is still attached to the same crane hook.
- 5.20.8 Crane hook safety latches, chains or devices shall be in good operating condition and shall be latched or fastened before the lift is made.

5.21 Floor Person

- 5.21.1 Floor personnel must ensure that no one is in a hazardous position before signaling a lift to be moved and must prevent personnel from entering the hazard area during movement of the lift.
- 5.21.2 Tag lines, hand hooks or equivalent "hands free" tools must be used as directed by the department for guiding and positioning lifts. Keep all parts of the body in the clear of lifts while tension is being applied. Never use your feet to hold cables or chains in place while the hoist is being raised or lowered or to steady or guide lifts.
- 5.21.3 Floor personnel shall ensure that lifts or lifting devices are set in such a way to prevent movement when unhooked from the crane. Stands, racks, blocking or other devices shall be used to assure the stability of the lift or lifting device.
- 5.21.4 Riding crane loads, lifting equipment or crane hooks by employees is prohibited.
- 5.21.5 Never walk, stand, or work under loads suspended from cranes. Maintain a safe distance from suspended loads.
- 5.21.6 Do not walk in the path of a moving crane load.

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- 5.21.7 Obey all crane status lights. Employees working in areas with overhead crane activity shall be trained to understand crane status lights.



- 5.19.8 *Sites can establish and train personnel on their status light colors, however, the site standard for status lights must include the configuration that magnet light OFF indicates power is on.*

5.22 Work Group Protection

- 5.22.1 Crane Interference Form (CCS_HHS_FRM_14.1) shall be initiated by a trained and Competent Cleveland-Cliffs management or designee and communicated with all affected employees prior to the commencement of the job requiring the Crane Interference Form and at the start of each subsequent shifts impacted by the work.
- 5.22.2 Crane movements may present special hazards. Other safe working procedures for the job being done must be followed while working on cranes or crane runways. This includes work on buildings, on top of tall bins, high machinery, and other areas where employees may be struck by the crane.
- 5.22.3 Additional precautions and safety measures to protect personnel when:
- a. Work, including installation of rail stops or hot rail grounds, is going to be performed on the crane runway including supporting structures above and below the crane rails that are within the crane envelope.
 - b. Work, including installation of rail stops or hot rail grounds, is going to be performed on a crane whenever there are two or more cranes on the same runway.
 - c. Work, including installation of rail stops or hot rail grounds, is going to be performed on the crane runway including supporting structures above and below the crane rails that does not have a permanent barrier between the immediately adjacent crane runways. A Crane Interference Form is required for both affected crane runways.
 - d. Mobile equipment with booms, man baskets, elevated work platforms or other attachments are being used within the building where overhead cranes are operating.

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5.22.4 Crane Interference Form shall be enacted for the following conditions:

- a. The mobile equipment in the process of performing work within the building is impinging or capable of impinging on the overhead crane's envelope.
- b. Note: Mobile equipment booms, man baskets, elevated work platforms or other attachments that are only traveling within the building with overhead cranes, and the boom or other attachments are in travel position, a Crane Interference Form is not required.
- b. If the fixed work locations, platforms, ladders, or temporary platforms within the building where overhead cranes are operating, and they impinge into the crane envelope, a Crane Interference Form is required.
- c. Fixed work locations and platforms that require a Crane Interference Form to access must have proper signage and gates at all entrance points that, at minimum, state what specific procedures and a completed Crane Interference Form evaluation and sign-off sheet (CCS_HHS_FRM_14.1 must be in place to access the platform.)
- d. Temporary work platforms actively in use such as scaffolding, that require a Crane Interference Form to be enacted to access, must be red tagged whenever a Crane Interference Form is not in place to access. The red tag must specify that a Crane Interference Form is required to access the platform.
- e. At the start of each turn or when erected or staged, a Crane Interference Form (CCS_HHS_FRM_14.1) shall be completed when the temporary work platform is actively in use but does not impinge into the crane envelope.
- f. The completed Crane Interference Form (CCS_HHS_FRM_14.1) shall be communicated with all affected crane operators prior to start of the shift or immediately after erecting or staging temporary work platforms.

5.22.5 Erected temporary work platforms that do not impinge into the crane envelope and are occupied shall be further highlighted by a flashing amber light. If the temporary work platform, mobile equipment, and ladders do not impinge and is not occupied, an amber light is not required.

- A crane shall proceed with caution past a flashing amber light, maintaining maximum distance as possible with lifting devices and suspended loads.

5.23 Protection Procedures

5.23.1 The means of Crane Interference Form protection shall be implemented in the order of effectiveness to eliminate the hazards of crane movement. The level of Crane Interference Form protection selected will be determined by a trained and competent manager or designee.

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5.24 Lock Out of Cranes

- 5.24.1 Locking out of overhead cranes shall be implemented when work activities, mobile equipment, work platforms, etc. enter or have the potential to impinge upon the crane envelope, unless infeasible. The decision to not lockout shall be made by a trained and competent manager or designee.
- 5.24.2 Lock out shall be completed for all cranes on the same runway.
- 5.24.3 Lock out shall be completed for all cranes on the adjacent runway if the crane movement presents a potential hazard to the work being performed on the adjacent crane, crane runway or in the crane envelope.
- 5.24.4 Lockout or installation of rail stops is required when work activity involves rail replacement, or work on rails where there is no safe clearance for cranes to pass.
- 5.24.5 When lock out has been determined by Cleveland-Cliffs Steel management as not feasible for the full length of the job, then positive rail stops shall be the next Crane Interference Form protective measure to be evaluated.

5.25 Positive Rail Stops

- 5.25.1 Crane lockout or rail stops are required when work activity involves rail breaks, rail replacement, or work on rails where there is no safe clearance for cranes to pass.
- 5.25.2 Positive rail stops shall be placed on crane rails between the crew doing the work and the active cranes to effectively protect the work zone. A specified distance for the rail stop placement will be determined by the Cleveland-Cliffs Steel management.
- 5.25.3 Blue flashing lights shall be positioned below the rail stops in a position readily visible to the crane operators to provide adequate warning of the position of the rail stops.
- 5.25.4 A new Crane Interference Form (CCS_HHS_FRM_14.1) shall be initiated by a trained and competent Cleveland-Cliffs Steel manager or designee and communicated with all affected employees prior to the commencement of the job requiring the Crane Interference Form and at the start of each subsequent shifts.
 - The previous Crane Interference Form (CCS_HHS_FRM_14.1) for LOTO and installation of the rail stops is in effect until the new Crane Interference Form (CCS_HHS_FRM_14.1) for the use of positive rail stops is completed and communicated with all affected employees.
 - At the commencement of the job requiring use of positive rail stop for Crane Interference, a new Crane Interference Form (CCS_HHS_FRM_14.1) shall be initiated.
- 5.25.5 Lock out with verification shall be completed for de-energization of the crane for the removal of the rail stops.

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- 5.25.6 Once the rail stops have been removed, the Crane Interference Form for the rail stops and the Crane Interference Form for de-energization of the crane for the removal of the rail stops can be cancelled.
- 5.25.7 When rail stops for Crane Interference protection has been determined by Cleveland-Cliffs Steel management or designee as not feasible for the full length of the job, then (a planned work zone shall be the next Crane Interference protective measure to be evaluated.

5.26 Planned Work Zones

- 5.26.1 The work group supervisor or leader shall conduct a meeting to determine the location of the safety lookouts(s) and blue flashing light(s).
- 5.26.2 Safety lookouts(s) and blue flashing light(s) shall be located at a minimum of 35 feet from the work zone. If the blue flashing light cannot be located at least 35 feet in advance of the planned work zone, it can be moved no closer than 10 feet from the planned work zone and the active crane(s) shall not be operated any faster than 2nd point within 2 bays approaching the blue flashing light. This shall be communicated to the affected crane operator(s).
- 5.26.3 The work group leader or supervisor shall then hold a Take 5 with the lookout to instruct them and position them as required. (Note: the meeting can be held with all parties in attendance as well).
- 5.26.4 When a job continues into the next turn, relief personnel shall have the same specific meeting to be advised of the details of the areas being protected.
- 5.26.5 Planned work zones with flashing warning beacons and safety lookout control provides 3 levels of control:
- First, the crane operator is responsible for the safe operation of the crane.
 - Second, blue flashing lights controlled by the work crew shall identify the planned work zone. The crane operator shall verify that the blue light is visible and that he has radio communication with the lookout.
 - Third, a safety lookout with a green hi-vis vest positioned by the crew, and working under the crew supervisor or leaders direction, shall constantly monitor crane activity for protection of the work crew. The safety lookout is not permitted to perform any other tasks.
- 5.26.6 There shall be always voice or radio communication between the safety lookout and the crane operator prior to the beginning of the work and. If a loss of communication occurs, the crane operator or lookout shall immediately notify the work crew(s) to stop, move equipment and personnel out of the travel path of the crane until communication can be re-established.
- 5.26.7 The safety lookout's sole duty shall be to keep a sharp lookout of crane activities for the crew being protected.

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- 5.26.8 The safety lookout will only be assigned to one crane.
- 5.26.9 When this method is used Crane Interference Form (CCS_HHS_FRM_14.1) shall be completed.
- 5.26.10 The safety lookout will hold the Crane Interference Form (CCS_HHS_FRM_14.1) until the job has been completed and the crew being protected relieves the safety lookout.
- 5.26.11 It is the responsibility of the trained and competent Cleveland-Cliffs Steel management or designee to be sure this Crane Interference Form (CCS_HHS_FRM_14.1) is completed and understood by all affected personnel.
- 5.26.12 In all cases involving the safety of the work crew, the crane operator shall follow the directives of the safety lookout person.
- 5.26.13 During work it may be necessary for crane(s) to pass into or through a planned work zone. The following shall be done:
 - a. The crane operator notifies the safety lookout and request permission to travel into or through the planned work zone.
 - b. The safety lookout will:
 - 1. Instruct the work crew to position all equipment out of the incoming crane's envelope.
 - 2. Move to a designated safe location which ensures the work crew will not be struck by the overhead crane as the crane travels into or through the planned work zone.
 - 3. Receive confirmation that the work crew has moved to the designated safe location and all pieces of equipment are outside of the incoming crane's envelope.
 - 4. Turn off the flashing blue light and communicate with the crane to pass or travel into the protected work zone.
 - 5. After the crane has exited out of the planned work zone, the blue flashing lights shall be reactivated.
 - 6. Contact the crane operators that the planned work zone is active again.
- 5.26.14 Riding outside the cab or landing platform is prohibited except for inspection and adjustment of parts, or where necessary in the performance of maintenance work. Riding on the trolley while the crane is in motion is never permitted.

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5.26.15 Persons checking the performance of the crane while it is in operation shall have:

- a. Department clearance accompanied with a written procedure signed by the maintenance shift supervisor.
- b. A clear understanding and communication with the crane operator of the position(s) of the service people.
- c. All other motions, other than the one being checked, shall be locked out, or the crane operator in the cab, in communication with the repair person, and instructed by the repair person.

5.26.16 A person must guide all travel of any Cleveland-Cliffs Steel owned or contractor owned mobile crane entering or operating, in any building, in which overhead cranes are operating.

5.27 Working on Cranes

5.27.1 Prior to commencing any preventative, break down or shutdown maintenance on an overhead crane, a Red Pennant Flag (highly visible) must be lowered from the crane 10 to 15 feet above the floor.

5.27.2 The pennant shall have "WARNING" displayed on both sides and shall easily be seen from the floor.

5.27.3 Warning signs shall also be placed on the floor (i.e., "Caution Overhead Work") to alert others in the area of the work being performed on the crane overhead.

5.27.4 The warning signs will be the maximum protection required if there is no potential for tools, parts, or materials to fall to the area below.

5.27.5 A Red Danger Tape Zone (with identification, i.e., department name, Contractor name) shall be erected to completely encompass and block access to work areas on the ground when there is a potential for tools, parts, or materials to fall from the crane during the maintenance activity.

5.27.6 Entry into the Red Zone shall be prohibited during the maintenance activity.

5.27.7 Entry into the Red Zone shall be prohibited until all conditions are controlled. Prior to Entry permission must be granted by work group.

6 Forms and Documented Information

6.1 Crane Interference Form - CCS_HHS_FRM_14.1

6.2 Crane Pre-use Inspection Checklist - CCS_HHS_FRM_13.2

7 Opportunities and Risks

7.1 This document provides the framework to reduce the likelihood of an overhead crane related injury.

7.2 The risks addressed in this standard are:

- Working at heights
- Control of hazardous energy
- Dropped objects
- Caught by/caught between hazards

8 Revision History

Rev.	Date	Summary of change	Authorized by
1	9-1-2022	Initial issue	K. D. Berard
2	9-22-2023	Changes are identified with a delta. Δ • Clarification of when a lift plan is needed. • Clarification that a crane trainee will need to pass a test. • Added Crane Interference Form to crane training. • Changed the crane status light color requirement to have sites establish a plant standard light system. Require training on the site's standard colors. • Added the requirement to remove a below the hook device prior to an upper limit test (pre-shift). • Added basic elements of periodic, annual, and cranes not in regular use inspection elements. • Comment regarding people with pacemakers not working in close proximity to strong magnetic fields. • Other minor editorial modifications.	Nathan Berry